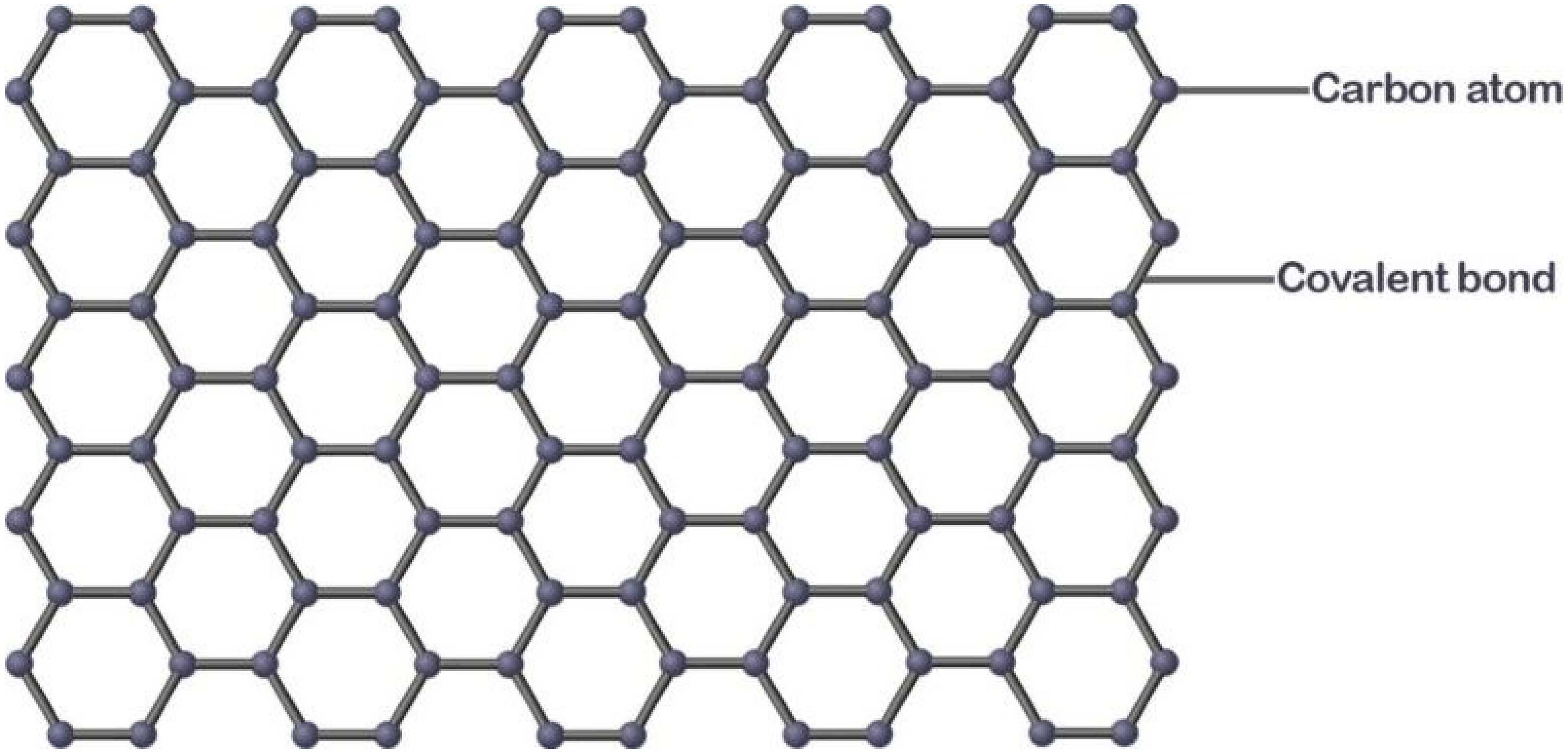
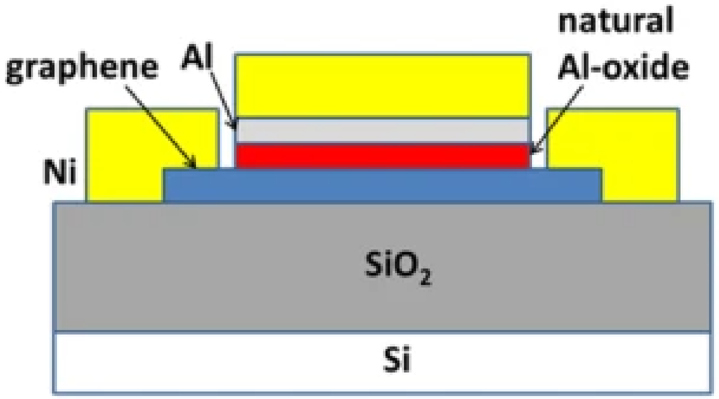
Beyond Silicon Devices: Graphene

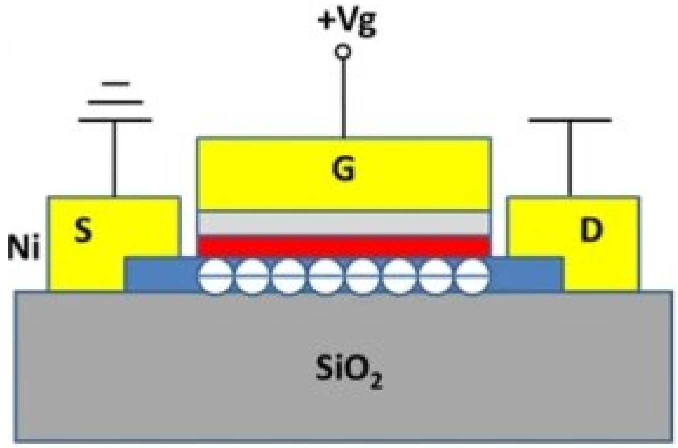


https://www.grapheneuses.org/wp-content/uploads/2020/05/Graphene.jpg

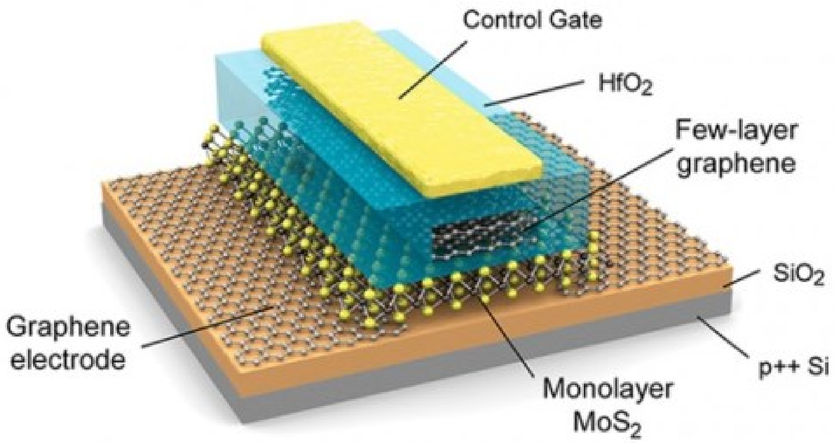
- Graphene is a single sheet of carbon atoms arranged in a crystal lattice (like a honeycomb pattern)

Beyond Silicon Devices: Graphene Transistors





Li, P., Zeng, R.Z., Liao, Y.B. et al. A Novel Graphene Metal Semi-Insulator Semiconductor Transistor and Its New Super-Low Power Mechanism. Sci Rep 9, 3642 (2019).



http://www.graphene-uses.com/wp-content/uploads/2015/10/Graphene-transistor-construction-details-e1444636284729.png

- Graphene transistors

Beyond Silicon Devices: Graphene Transistors

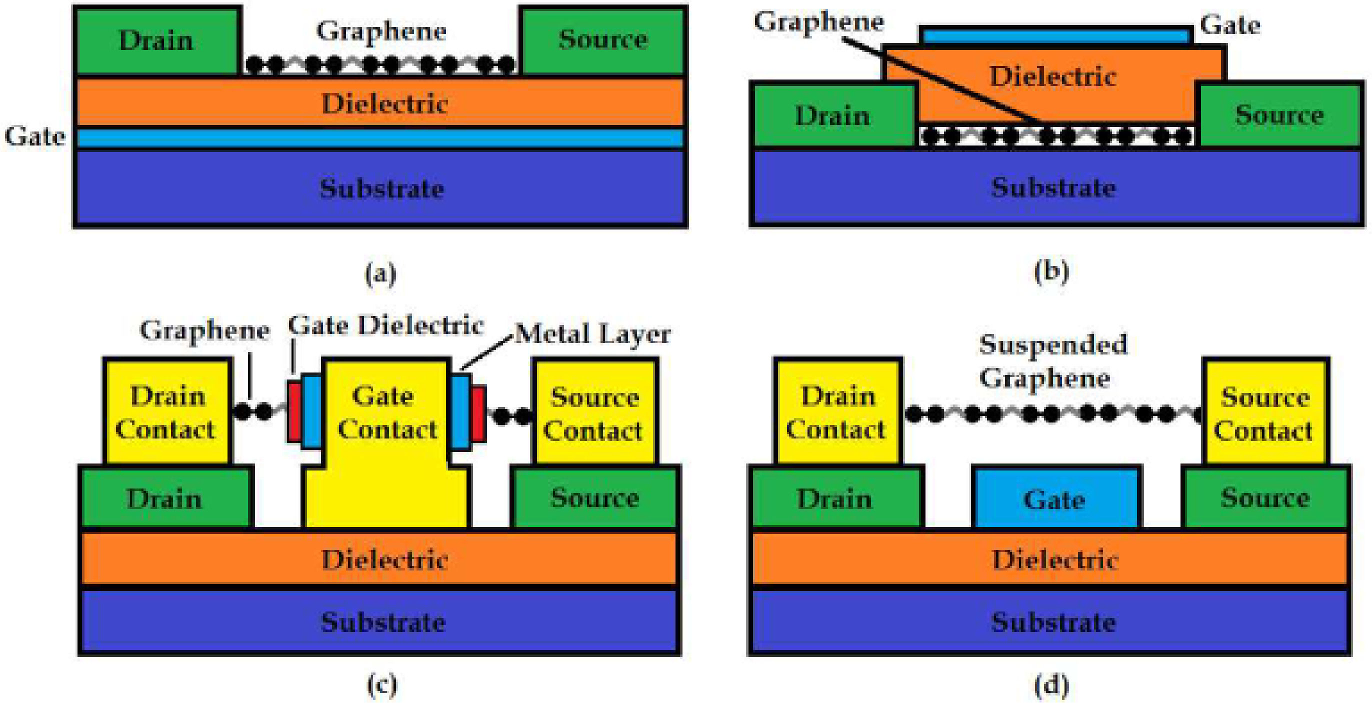
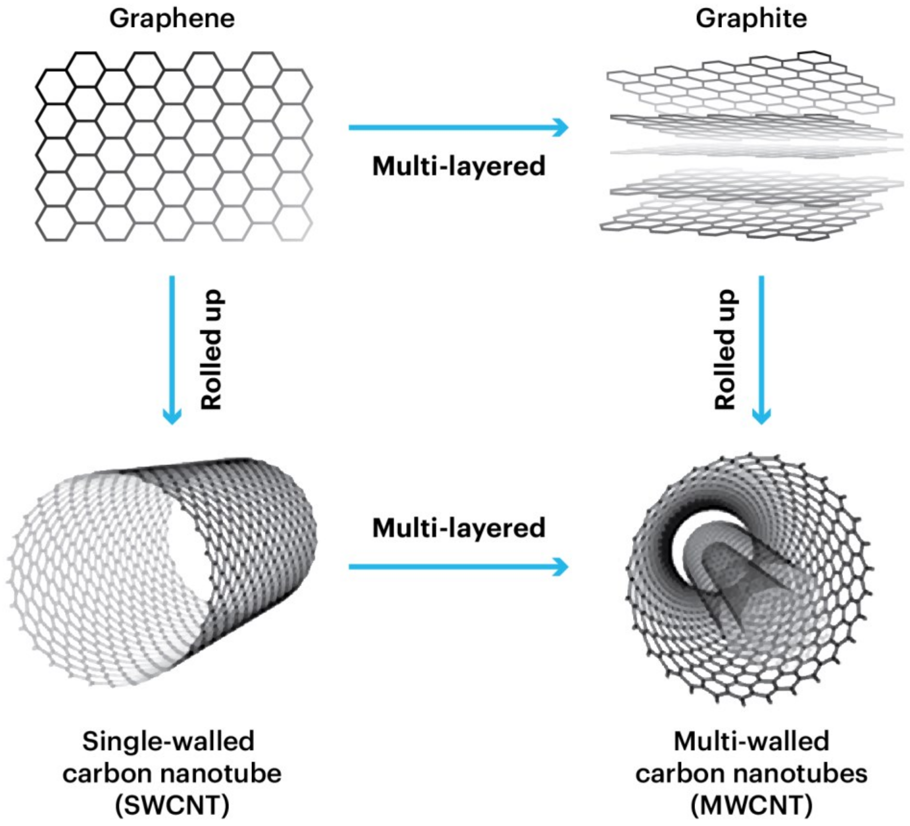


Fig. 9. Cross sections of different geometries of graphene field-effect transistors: (a) back-gated GFETs, (b) top-gated GFETs, (c) wrap-around gate GFETs, and (d) suspended GFETs.

R. Vargas-Bernal and G. Herrera-Prez, 'Carbon Nanotube- and Graphene Based Devices, Circuits and Sensors for VLSI Design', VLSI Design. InTech, Jan. 20, 2012.

- Graphene transistors

Beyond Silicon Devices: Carbon Nanotubes



https://tuball.com/media/imperavi_redactor_content/e7/b1/e7b13bed-6a23-4c89-acde-67c53fd3c775/%D0%BA%D0%B0%D1%80%D1%82%D0%B8%D0%BD%D0%BA%D0%B8-06%20(1).png

- Carbon nanotubes are nanoscale hollow tubes composed of carbon atoms

- Single-walled or multi-walled

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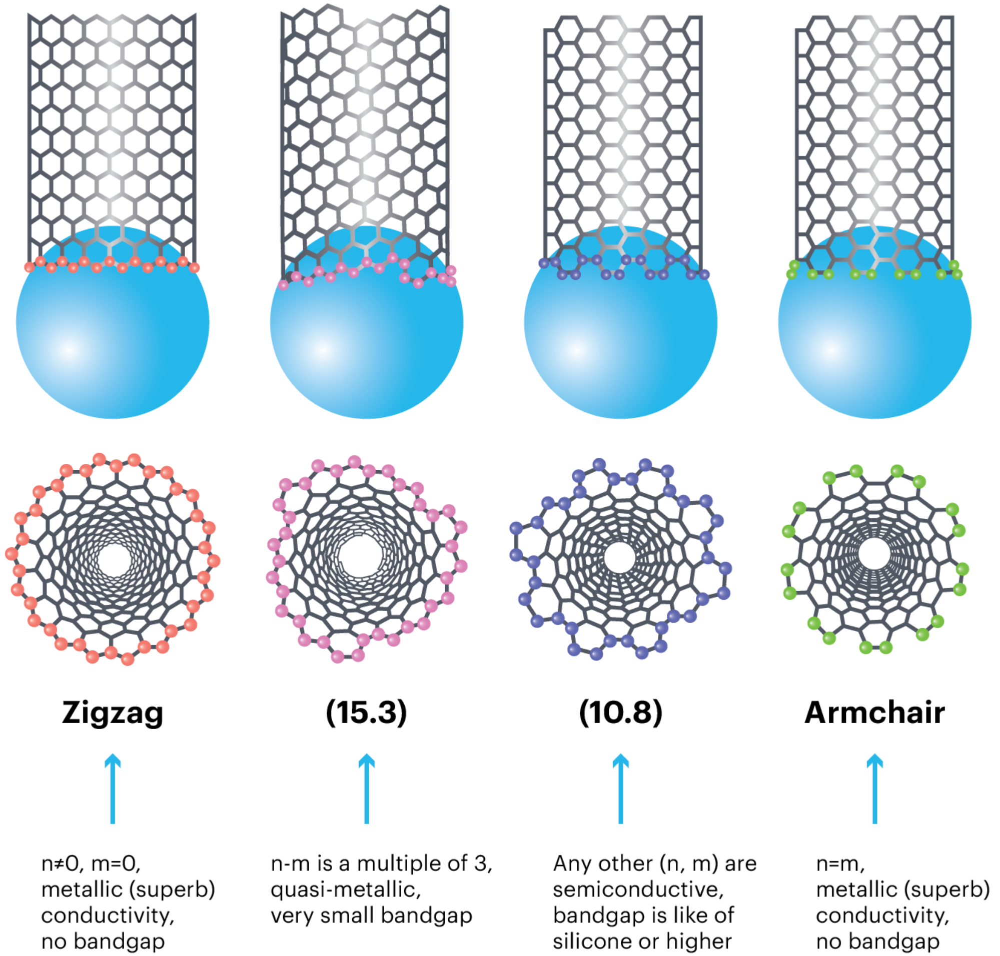


Figure 4. *The difference in chirality of SWCNTs*

https://tuball.com/media/imperavi_redactor_content/8c/fb/8cfbe1e7-b073-49e7-bf32-c98b1ce227e8/%D0%BA%D0%B0%D1%80%D1%82%D0%B8%D0%BD%D0%BA%D0%B8-01%20(2).png

- Carbon nanotubes are nanoscale hollow tubes composed of carbon atoms
- The internal orientation (chirality) of carbon nanotubes leads to different properties, as it leads to different bandgaps

Beyond Silicon Devices: Carbon Nanotubes

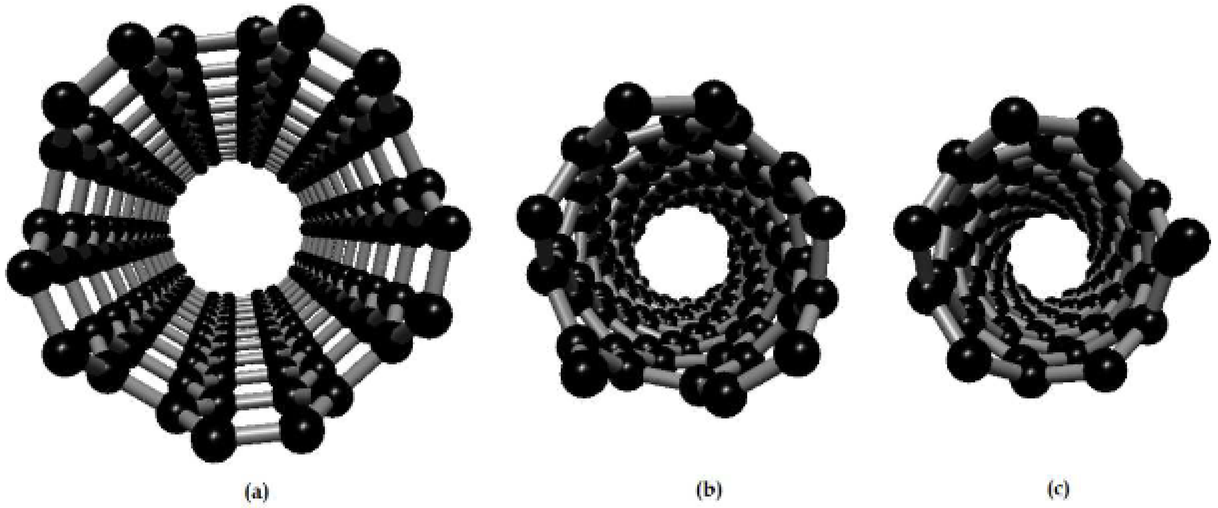


Fig. 3. Classification of carbon nanotubes by electrical properties: (a) metallic nanotube, (b) semiconducting nanotube, and (c) moderate semiconducting nanotube.

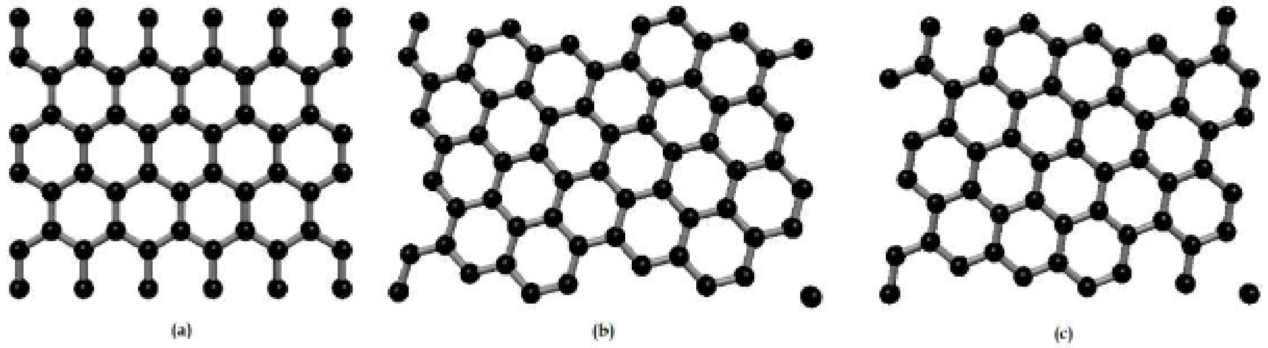
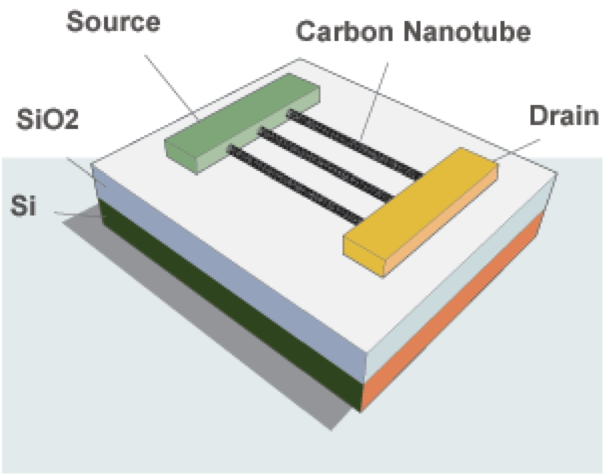


Fig. 6. Classification of graphene by electrical properties: (a) metallic graphene, (b) semiconducting graphene, and (c) moderate semiconducting graphene.

R. Vargas-Bernal and G. Herrera-Prez, 'Carbon Nanotube- and Graphene Based Devices, Circuits and Sensors for VLSI Design', VLSI Design. InTech, Jan. 20, 2012.

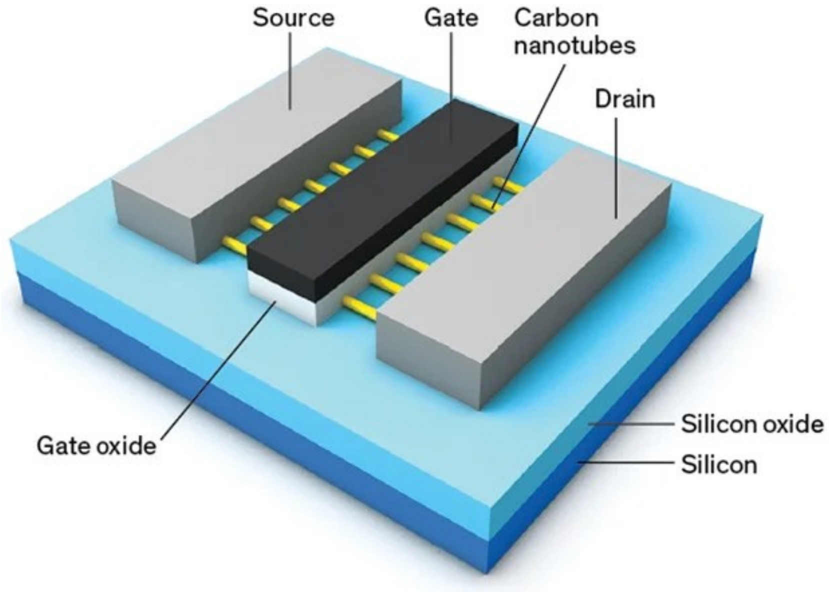
- Carbon nanotubes are nanoscale hollow tubes composed of carbon atoms
- The internal orientation (chirality) of carbon nanotubes leads to different properties, as it leads to different bandgaps

Beyond Silicon Devices: Carbon-Nanotube Transistors



https://nanointegris.com/wp-content/uploads/2017/08/Img-Transitors.png

- Carbon-nanotube transistors



https://spectrum.ieee.org/media-library/img-07-cntc-f2.jpg?id=25580940

Beyond Silicon Devices: Carbon-Nanotube Transistors

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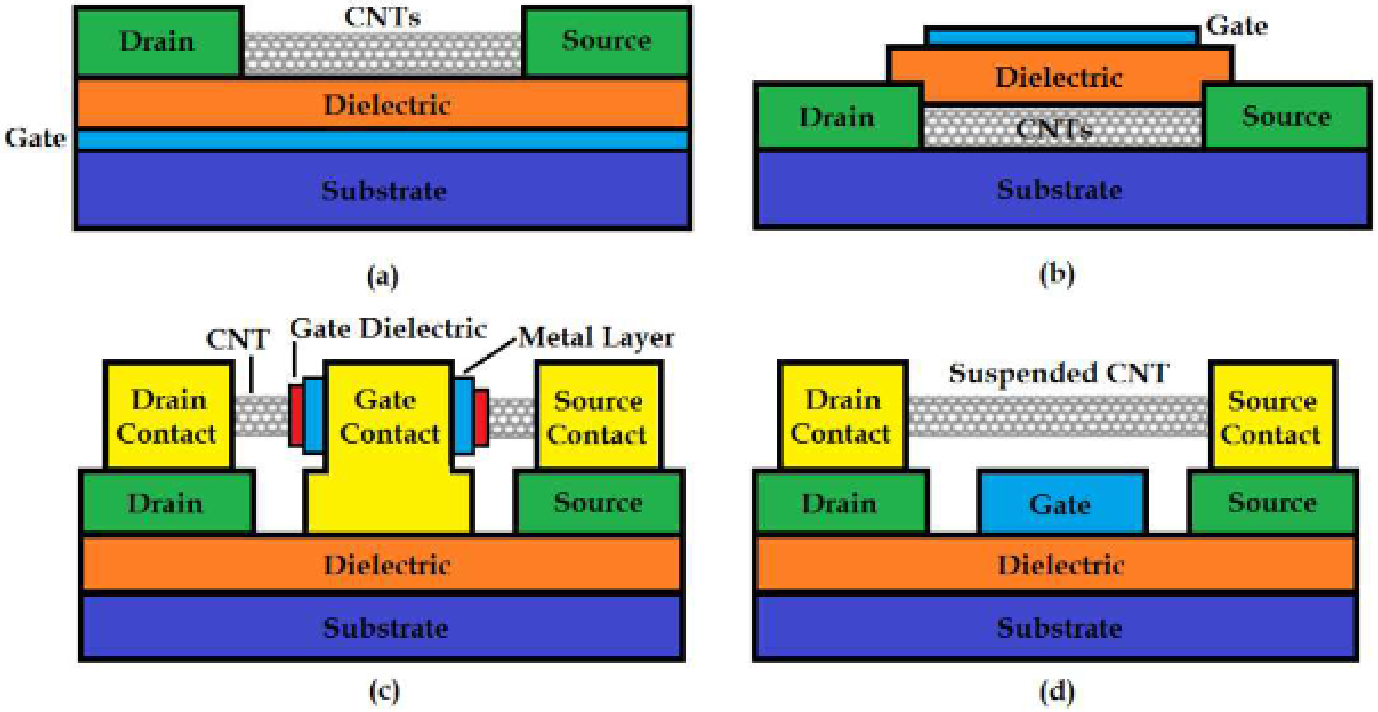


Fig. 8. Cross sections of different geometries of carbon nanotube field-effect transistors: (a) back-gated CNTFETs, (b) top-gated CNTFETs, (c) wrap-around gate CNTFETs, and (d) suspended CNTFETs.

R. Vargas-Bernal and G. Herrera-Prez, 'Carbon Nanotube- and Graphene Based Devices, Circuits and Sensors for VLSI Design', VLSI Design. InTech, Jan. 20, 2012.

- Carbon-nanotube transistors

Beyond Silicon Devices: Transistor applications

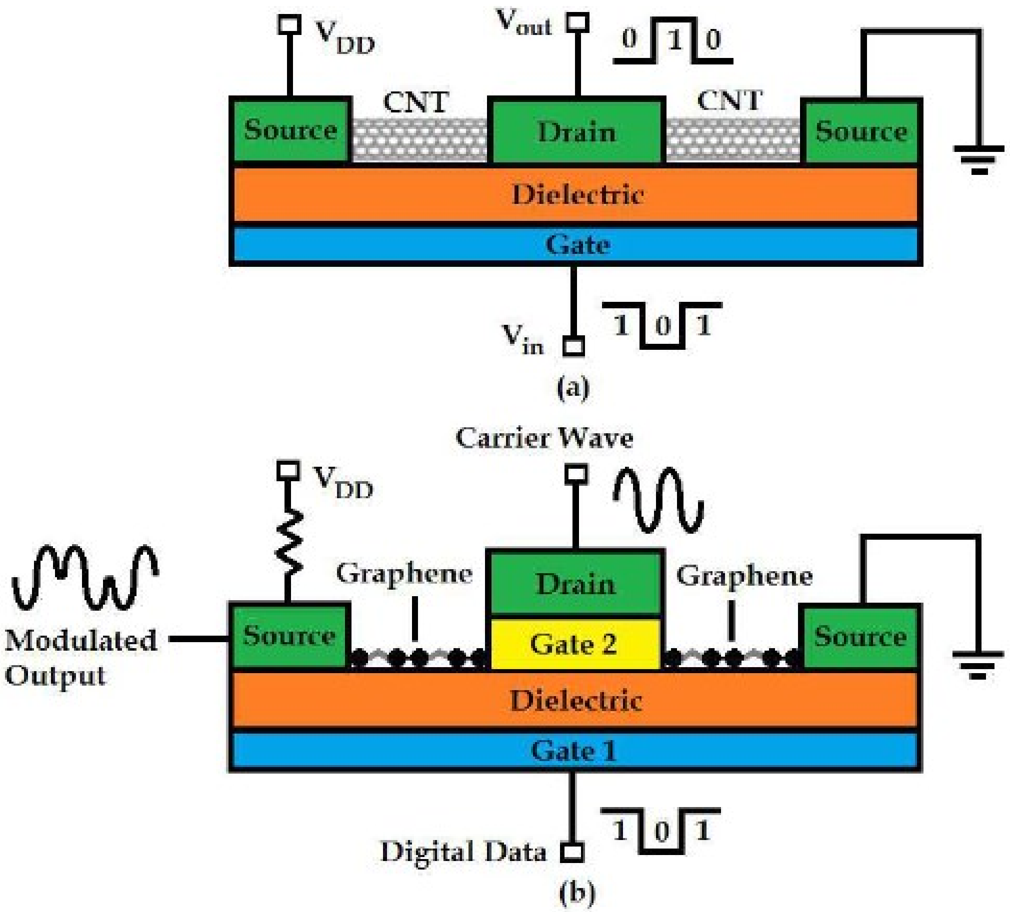
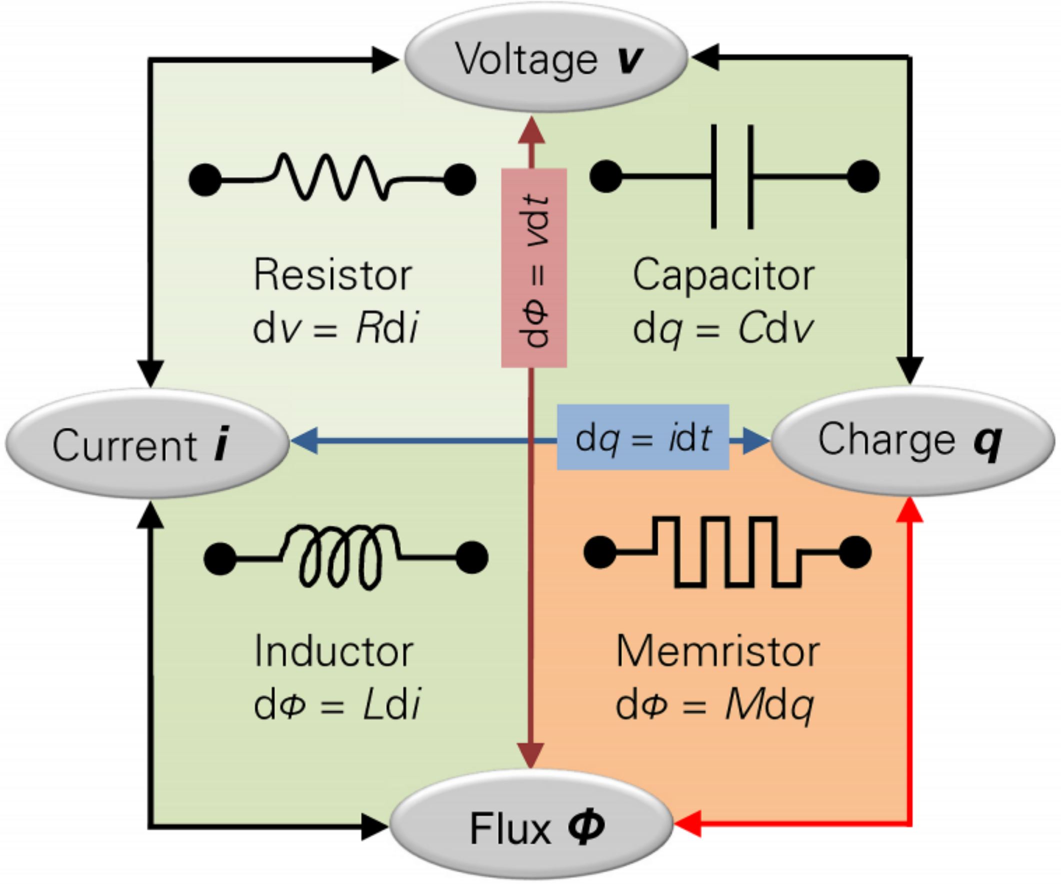


Fig. 10. VLSI circuits: (a) Carbon nanotube-based logic inverter, and (b) graphene-based digital modulator.

R. Vargas-Bernal and G. Herrera-Prez, 'Carbon Nanotube- and Graphene Based Devices, Circuits and Sensors for VLSI Design', VLSI Design. InTech, Jan. 20, 2012.

Beyond Silicon Devices: Memristors

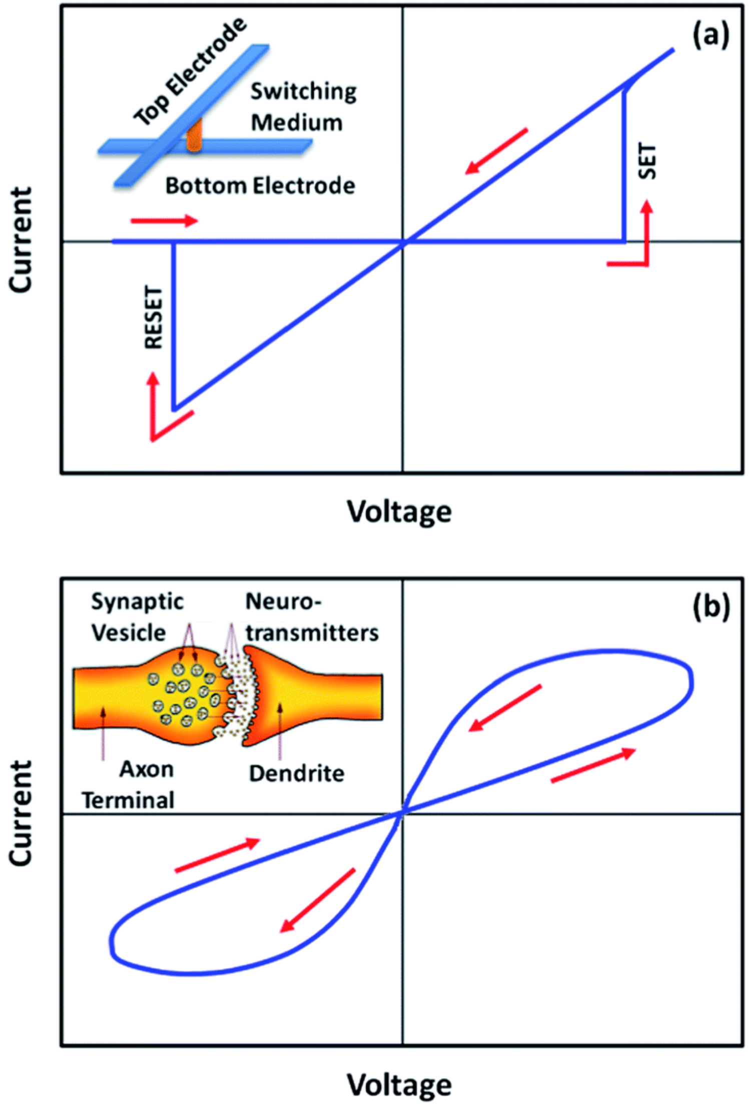


https://www.elprocus.com/wp-content/uploads/Memristor-768x644.png

- Memristors, from ‘memory’ and ‘resistor’, exhibit a resistive switching behaviour.

https://commons.wikimedia.org/wiki/File:Memristor-Symbol.svg

Beyond Silicon Devices: Memristors



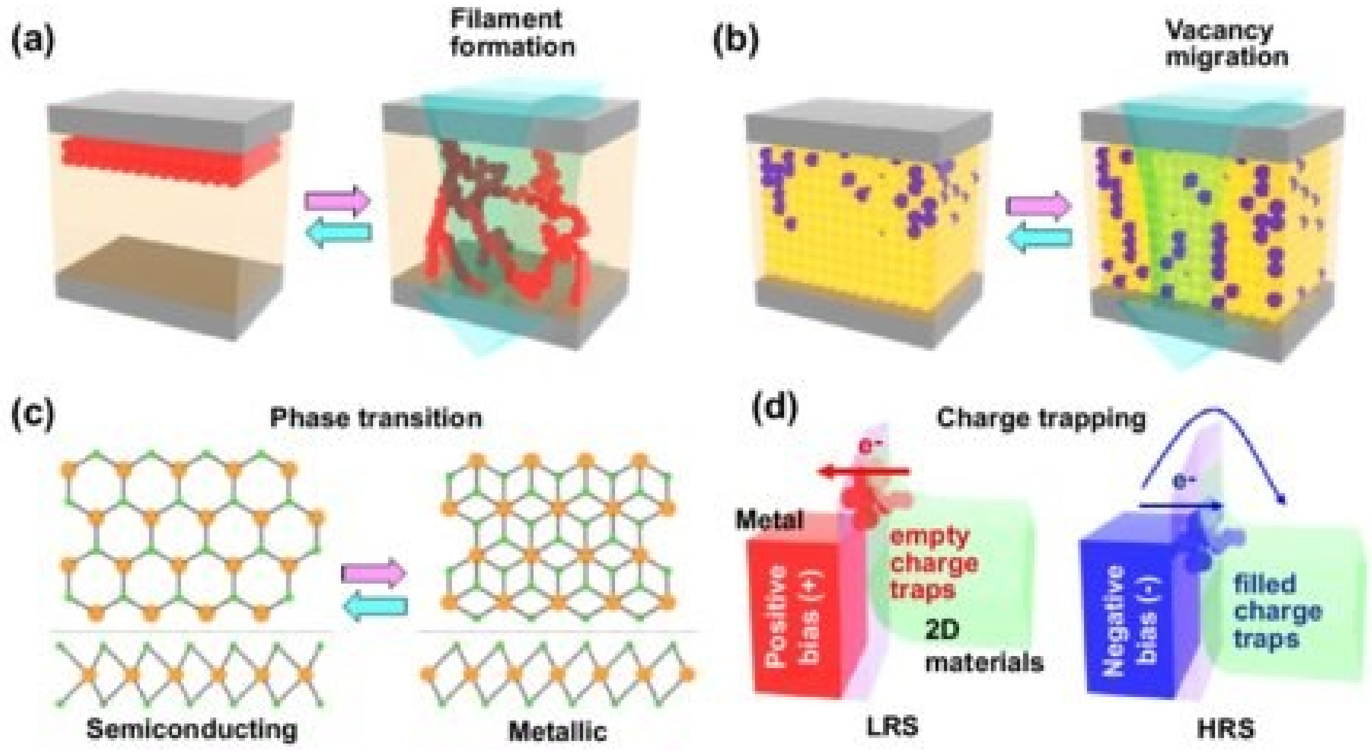
Chen, Y., Liu, G., Wang, C., Zhang, W., Li, R.-W., & Wang, L. (2014). Polymer memristor for information storage and neuromorphic applications. Materials Horizons, 1(5), 489–506.

- Memristors, from ‘memory’ and ‘resistor’, exhibit a resistive switching behaviour.

- Current increases abruptly when a certain voltage threshold is exceeded, resulting in a reduction in their internal resistance. The memristor is then in a state known as the ‘Low Resistive State’ (LRS).

- Conversely, if a voltage with reversed polarity is applied, the internal resistance of the device suddenly increases after exceeding a certain voltage threshold. In this case, the memristor is in a state called ‘High Resistive State’ (HRS).

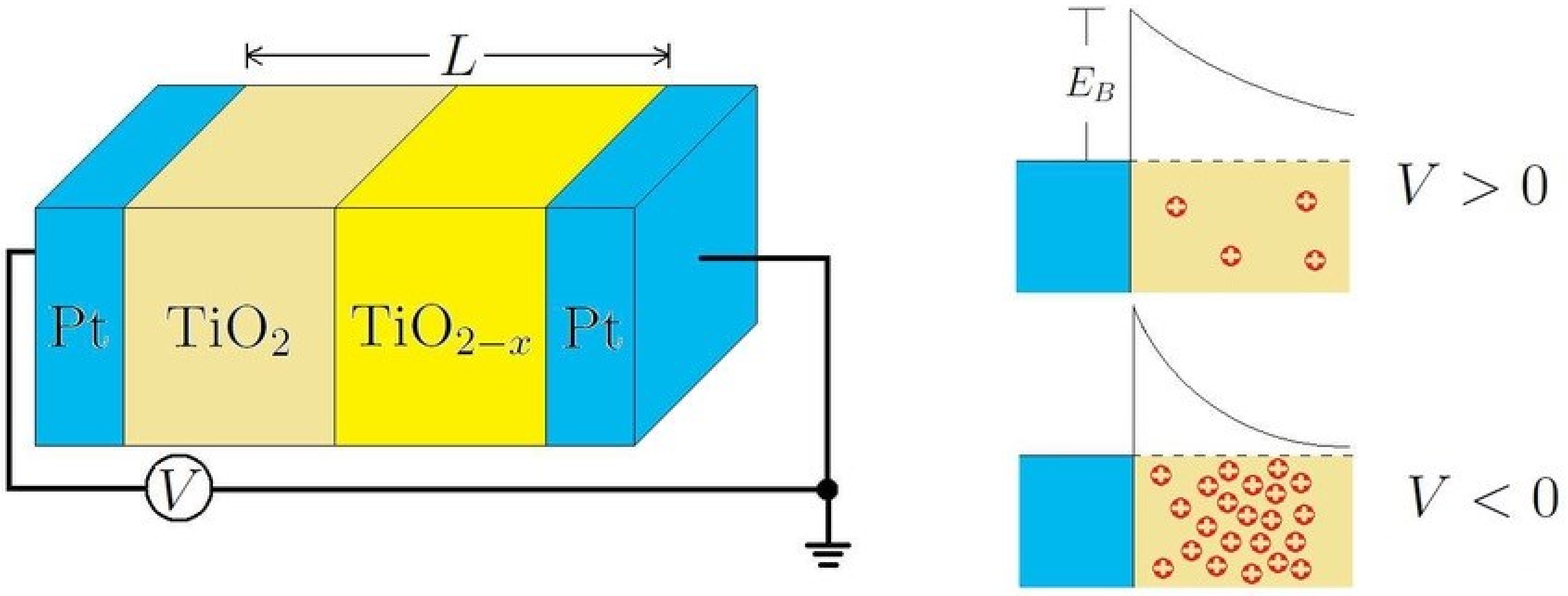
Beyond Silicon Devices: Memristor Materials



Kwon, K.C., Baek, J.H., Hong, K. *et al.* Memristive Devices Based on Two-Dimensional Transition Metal Chalcogenides for Neuromorphic Computing. *Nano-Micro Lett.* 14, 58 (2022).

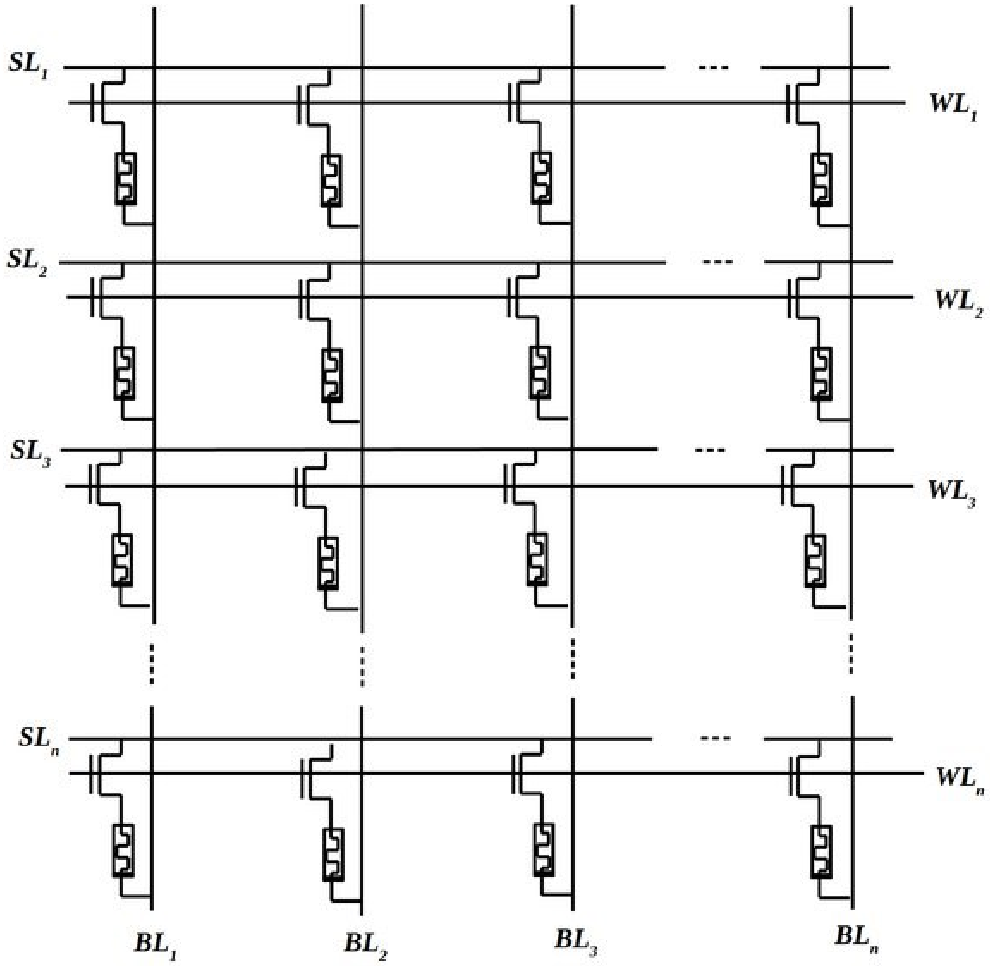
- Representative working principles of memristor devices

Beyond Silicon Devices: Memristor Materials



https://www.elprocus.com/wp-content/uploads/Construction-of-Memristor.jpg

Beyond Silicon Devices: Memristors as Memories



- Resistive Random-Access Memory (ReRAM or RRAM)

M. B. Majumder, M. S. Hasan, M. Uddin and G. S. Rose, "A Secure Integrity Checking System for Nanoelectronic Resistive RAM," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 27, no. 2, pp. 416-429, Feb. 2019.

The Future of Silicon-Based Transistors

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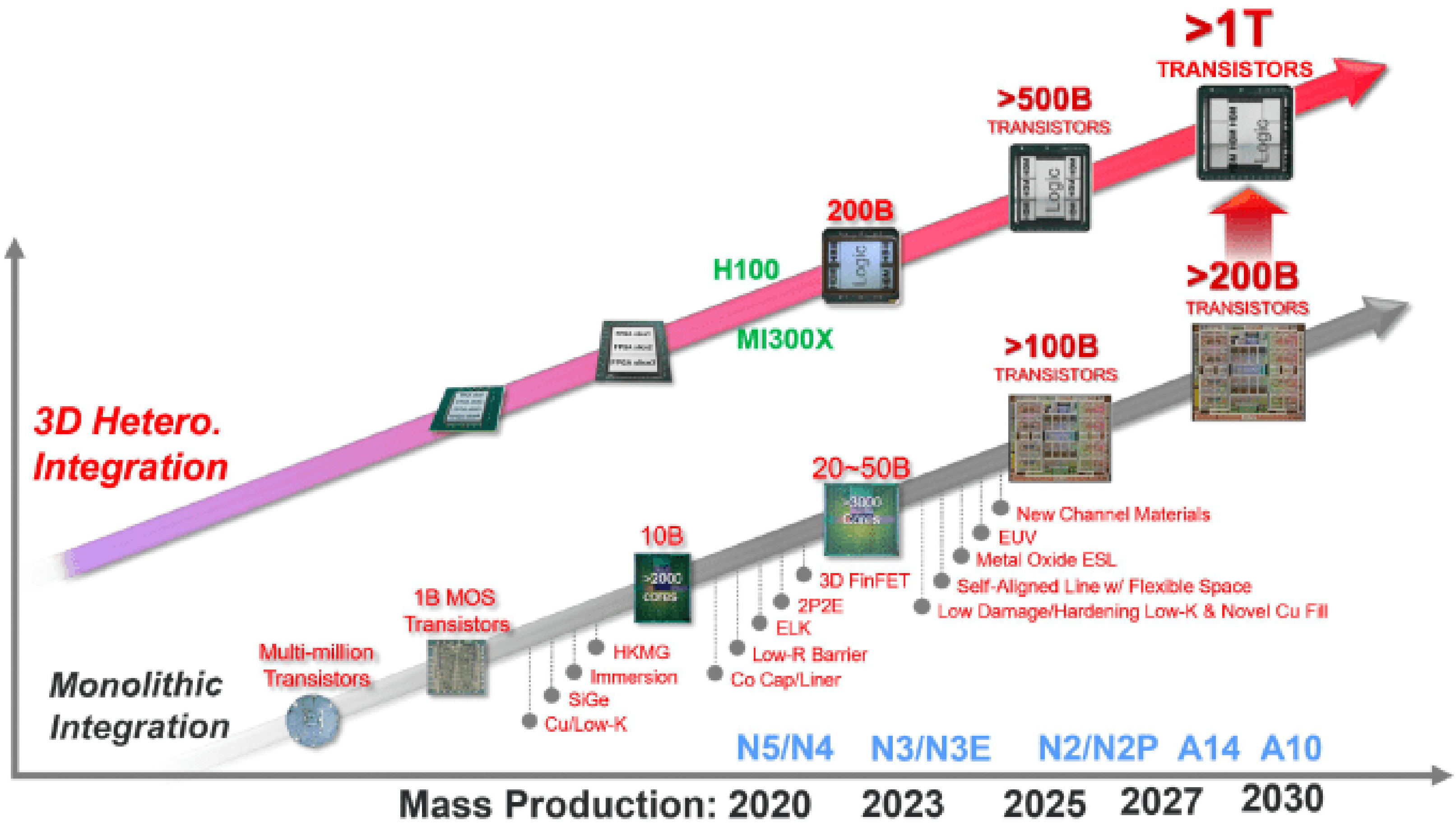

https://www.hardwareluxx.de/images/cdn02/uploads/2024/Jan/ample_heatsink_f6/tsmc-1nm-roadmap_680px.png

The Future of Silicon-Based Transistors

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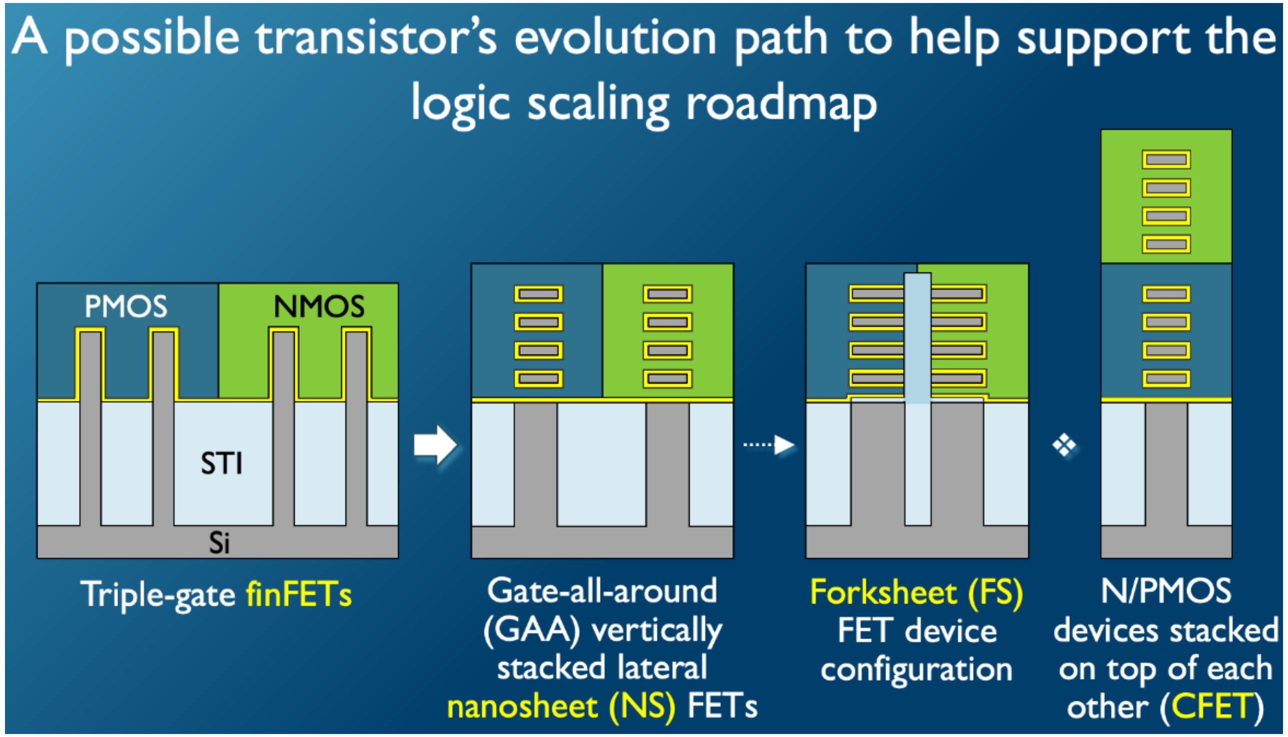

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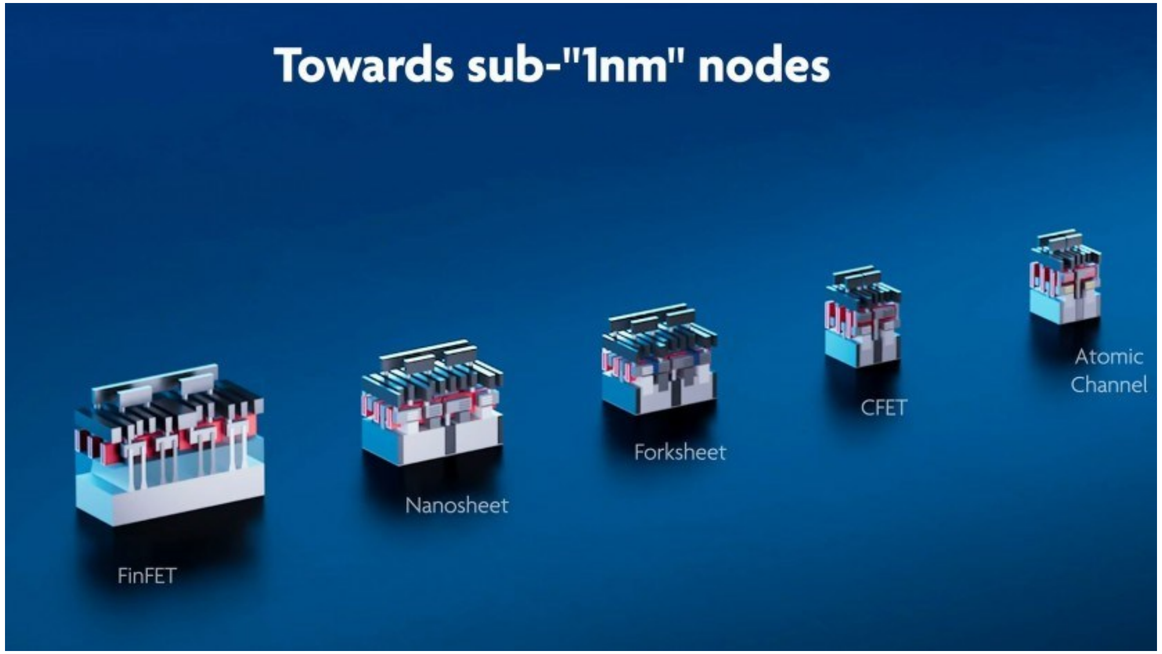

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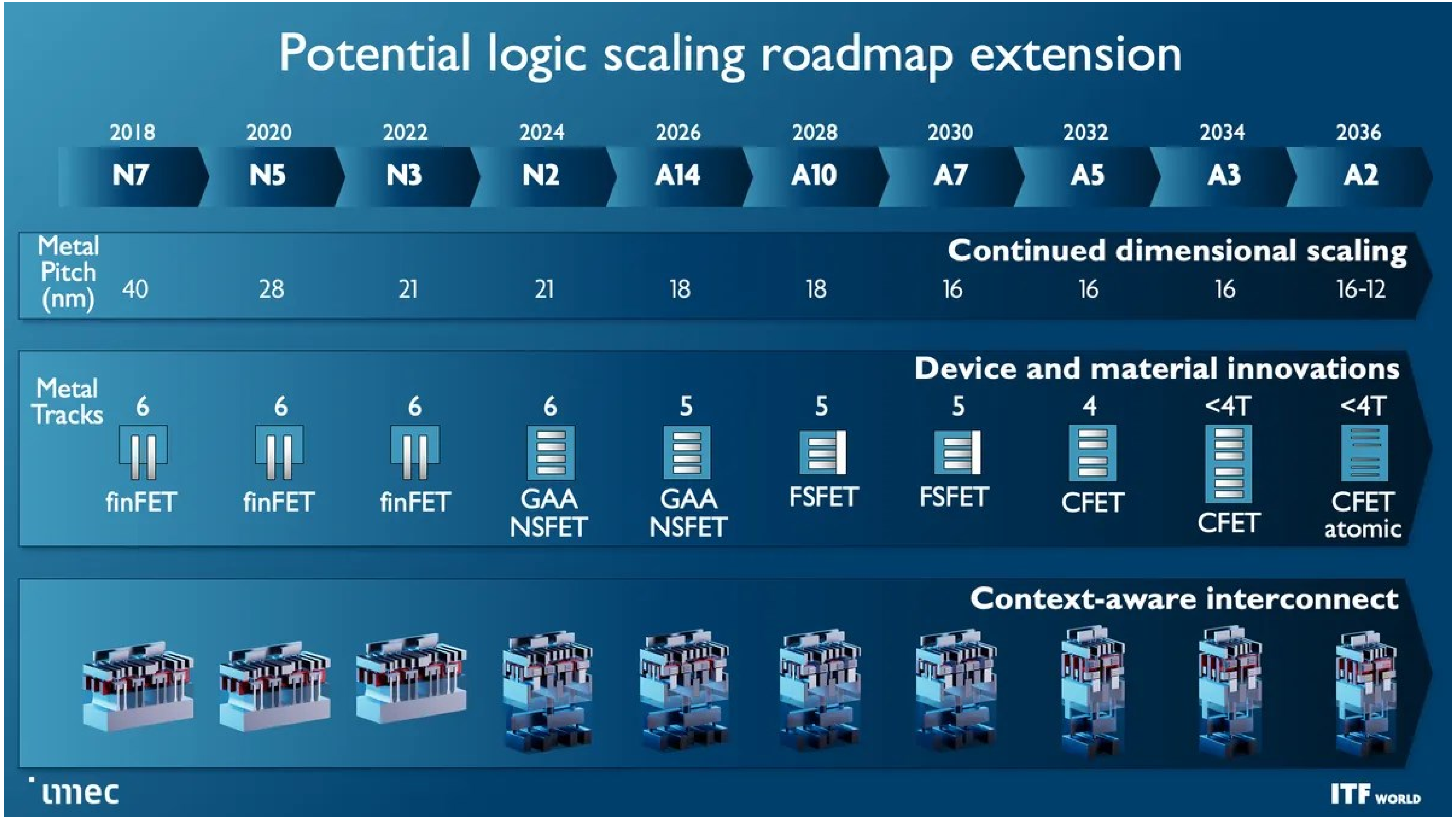
The Future of Silicon-Based Transistors



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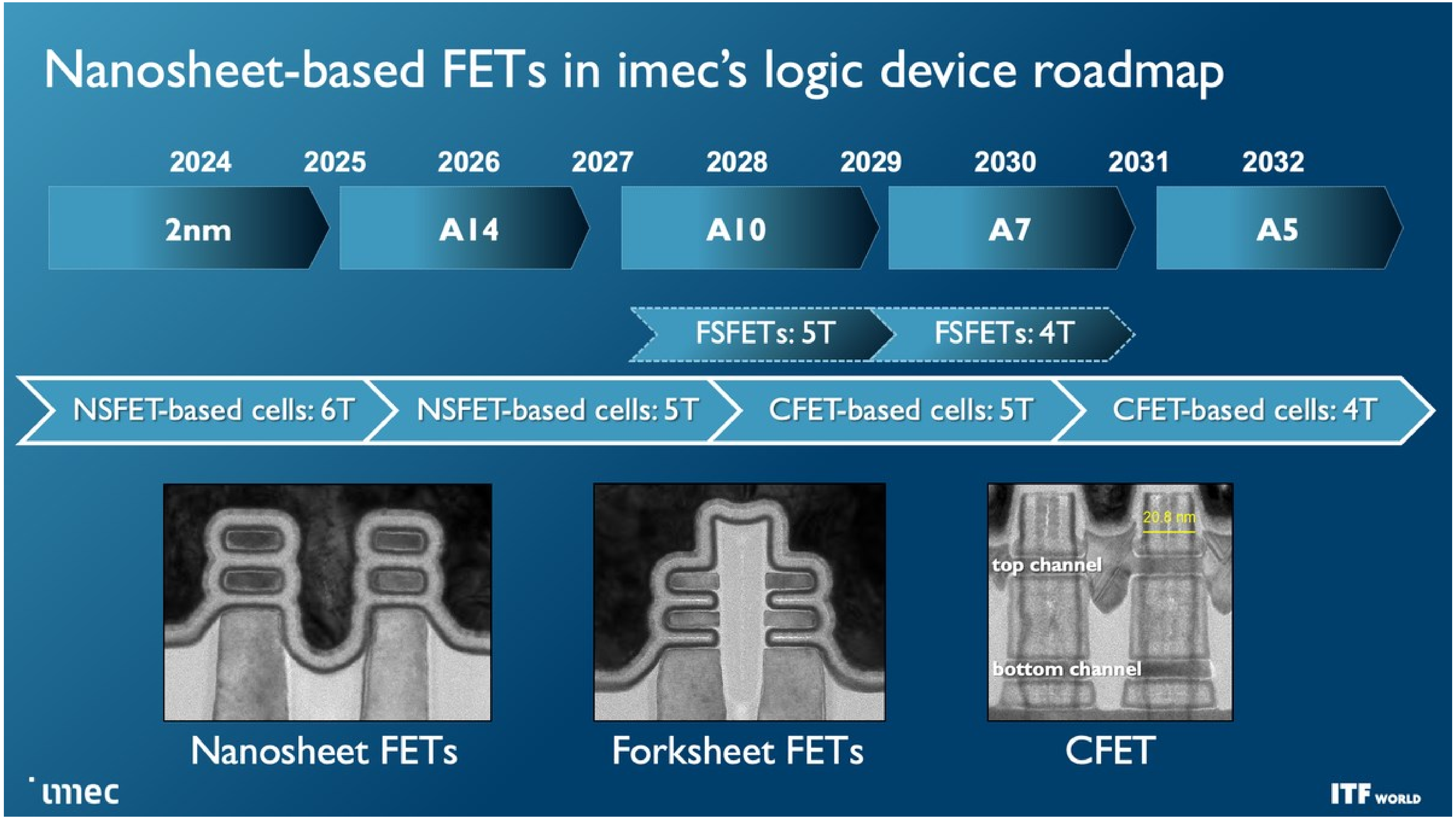
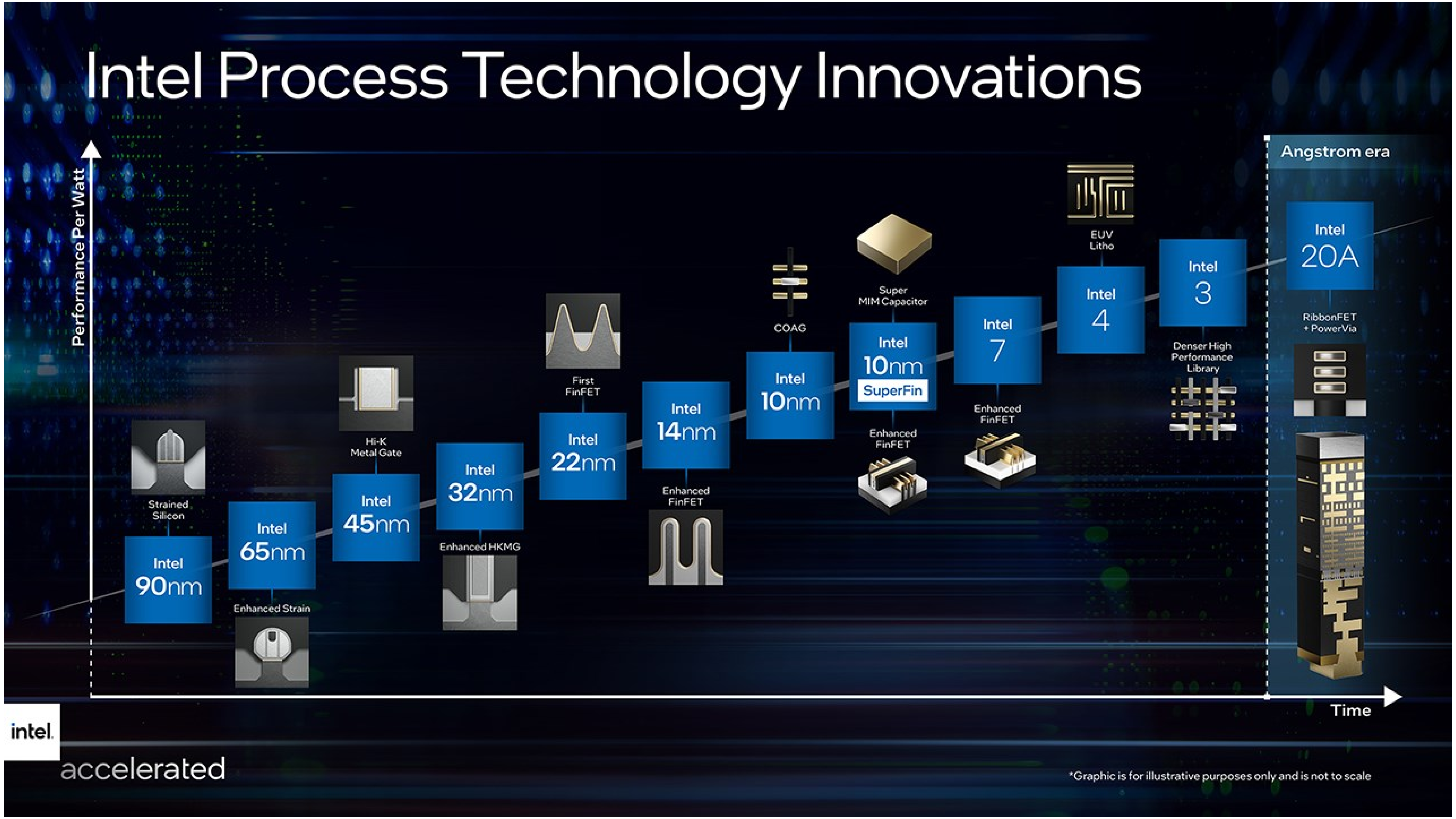
The Future of Silicon-Based Transistors



https://i0.wp.com/www.fanaticalfuturist.com/wp-content/uploads/2024/02/imec3.png?w=1200&ssl=1

The Future of Silicon-Based Transistors



https://cdn.mobilesyrup.com/wp-content/uploads/2021/07/intel-road-map.png

Circuit Design With the Help of Artificial Intelligence

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- SnapEDA's groundbreaking tool, SnapMagic Copilot
  - It enables the use of AI algorithms for a more streamlined creation process.
  - The synergy between generative AI and other advanced manufacturing technologies, such as additive manufacturing, beckons a future replete with highly customized products.

  - **Auto-completion of circuits:** Drop a microcontroller onto a circuit, and SnapMagic Copilot effortlessly places the suggested decoupling capacitors for you.
  - **Natural language for designs:** Instruct your PCB tool in plain English, like requesting a 'non-inverting amplifier with a gain of 2', and SnapMagic Copilot will craft those circuits using orderable components based on solid circuit theory fundamentals.
  - **Reference design recommendations:** Upon integrating a relevant integrated circuit, SnapMagic Copilot proactively proposes a manufacturer-endorsed reference design.
  - **Optimization driven by AI:** Want your bill-of-materials optimized by cost? We'll suggest pin compatible replacements that will suit your needs.
  - **Smart component replacements:** When SnapMagic Copilot spots a component running low on stock, it intelligently recommends alternatives using our specialized component recommendation system

https://techexplorations.com/blog/artificial-intelligence/ai-in-circuit-design/

https://blog.snapeda.com/2023/10/04/snapeda-is-now-snapmagic-the-ai-copilot-for-electronics-design/

Circuit Design With the Help of Artificial Intelligence

Faculty of Computer Science and Mathematics

- Autodesk's Artificial Intelligence (AI) and generative design tools for the Design and Make industries
  - Autodesk AI
  - AutoCAD
- NVIDIA Metropolis is an application framework, set of developer tools, and partner ecosystem that brings visual data and AI together to improve operational efficiency and safety across a broad range of industries.
- Companies building Automation(AI) tools for the Hardware / PCB Design
  - JITX (USA)
  - Contunity Now Celus (Germany)
  - Flux AI PCB Design Tool with Copilot (https://www.flux.ai/p)
  - CADY: AI-based Schematic Inspection Tool (https://cadysolutions.com/)
  - Circuit Tree (India)
  - CircuitMind (UK)
  - Zuken, PCB Design using AI (Japan)
  - InstaDeep (UK)
  - Gumstix Geppetto (USA)

https://www.autodesk.com/products/autocad/overview

https://www.nvidia.com/en-us/autonomous-machines/intelligent-video-analytics-platform/

https://pallavaggarwal.in/automated-pcb-design-using-ai/